

ADV-Lens — User Manual

A printable desk reference for compliance officers, M&A diligence teams, and the engineers who run the system

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1. What this is, in plain English

ADV-Lens reads a registered investment adviser's regulatory disclosure brochure (the SEC Form ADV Part 2A — see the vocabulary section if any of that is unfamiliar) and produces a structured **redline report** that a Chief Compliance Officer can use to:

- spot disclosure weaknesses against SEC plain-English expectations,
- compare the firm's fees, conflicts, and disciplinary disclosures to a cohort of peer brochures of similar size and strategy,
- have a defensible audit trail of what the system extracted, what it flagged, and which human approved or rejected the report.

It is **not** legal advice. It is **not** an auto-publisher. It is **not** a substitute for CCO judgment. Every report leaves the system only after a human review gate writes a row to an audit table.

The intended day-to-day reader is a CCO at a mid-market RIA, an M&A diligence analyst evaluating a target adviser, or a wealth-management consulting team reviewing a client's filing posture.

2. The vocabulary

Every piece of jargon that appears in this manual or in the system, in one sentence each. Skim, then come back when you hit a term you don't recognise.

2.1 Acronyms

Acronym	Stands for	What it means in this system
ADV	(no expansion — the SEC's form name)	The registration form every SEC- or state-registered investment adviser files. Part 2A is the public disclosure brochure ADV-Lens reads.
RIA	Registered Investment Adviser	A firm that gives investment advice for compensation and is registered with the SEC or a state regulator.
CCO	Chief Compliance Officer	The named person at an RIA accountable for compliance with the Advisers Act. The intended human user of ADV-Lens.
CRD	Central Registration Depository	The unique numeric identifier the SEC and FINRA use for every adviser firm and individual. ADV-Lens keys everything off the firm's CRD.
IARD	Investment Adviser Registration Depository	The SEC system advisers file ADV through. The bulk download contains every active adviser's Part 1 record.
IAPD	Investment Adviser Public Disclosure	The public-facing SEC website that publishes every adviser's brochures. ADV-Lens fetches PDFs from <code>files.adviserinfo.sec.gov</code> .
AUM	Assets Under Management	Total client assets the firm manages. ADV-Lens uses an AUM band (0–1B, \$1–5B, \$5–50B, \$50B+) as a peer-cohort filter.
SEC	Securities and Exchange Commission	Federal regulator. Registers and examines RIAs above \$100M AUM (state regulators handle smaller).
FINRA	Financial Industry Regulatory Authority	Self-regulator for broker-dealers; runs BrokerCheck, the public lookup for adviser/broker disciplinary history.
HITL	Human-In-The-Loop	A pipeline pattern where machine output is gated on a human approval step before being released. ADV-Lens's <code>HumanReviewGate</code> node implements this.
RAG	Retrieval-Augmented Generation	Pulling relevant text from a knowledge store and feeding it to an LLM as context. ADV-Lens uses RAG to put peer-brochure excerpts in front of the redline writer.
BRCHR_VRSN_ID	Brochure Version ID	The SEC's per-version identifier for a specific filing of a brochure. The same firm can have multiple brochures (e.g., wrap fee program disclosure plus main brochure).
ERA	Exempt Reporting Adviser	A firm that meets a registration exemption but still files an abbreviated ADV. ADV-Lens does not currently process ERAs; it targets full Part 2A filings.
M&A	Mergers and Acquisitions	One target use case: an acquirer's diligence team reading a target adviser's ADV.
ADR	Architecture Decision Record	A short markdown file in <code>docs/adr/</code> capturing one design decision and its consequences. Eleven exist today.

2.2 Industry jargon

Term	One-sentence definition
Fee schedule	The firm's published pricing — typically a tiered percentage of AUM, sometimes flat retainer or hourly, found in Item 5 of the ADV.
Conflict of interest	Any arrangement that gives the firm a financial incentive to act against a client's interest — affiliated funds, soft dollars, principal trading, etc. Disclosed in Items 10/11/12 .
Disciplinary disclosure	Any past regulatory, criminal, or civil action involving the firm or its principals. Disclosed in Item 9 . The CCO must keep this current.
Annual ADV amendment	Every adviser must file an updated ADV within 90 days of fiscal-year end. ADV-Lens sees the most recent filing on IAPD.
Wrap fee program	A bundled-fee program where one charge covers advice + execution + custody. Has its own brochure variant (Part 2A Appendix 1).
Custodian	The bank or broker-dealer that holds client assets. Disclosed in Item 12 when relevant.
Compliance exam	The SEC or state regulator's periodic on-site review of an adviser's books, records, and disclosures. ADV-Lens is designed to make exam prep cheaper, not to replace exam-readiness work.
Soft dollars	Research or services a custodian provides to the adviser in exchange for client trading commissions — disclosed in Item 12 and gated by the § 28(e) safe harbor .
Performance fee	Compensation tied to investment performance (e.g., 20% of gains). Permitted only with qualified clients; disclosed in Item 5.
Code of Ethics	A firm's written rules governing personal trading by access persons, required by Rule 204A-1, summarised in Item 11.

If a term in this manual is not in the table above, it appears in the parking-lot doc (`docs/parking-lot.md`) or the architecture doc as a candidate for inclusion.

3. What goes in: a Form ADV Part 2A brochure

The system's input is a single PDF — the firm's most recent Part 2A brochure as published on IAPD. Brochures are typically 30–80 pages and follow a fixed structure of 18 numbered Items the SEC mandates.

Item	Title	What's in it
1	Cover Page	Firm name, address, contact, brochure date.
2	Material Changes	Summary of changes since last annual amendment.
3	Table of Contents	Required structural element.
4	Advisory Business	Firm history, ownership, services, total AUM.
5	Fees and Compensation	Fee schedules, billing, third-party costs. Read by the Fee Extractor.
6	Performance-Based Fees	Disclosure if any.
7	Types of Clients	Individuals, pensions, funds, etc.
8	Methods of Analysis	Investment process.
9	Disciplinary Information	Regulatory, criminal, civil actions. Read by the Disciplinary Extractor.
10	Other Financial Industry Activities and Affiliations	Affiliated B/D, insurance, fund complex, bank. Read by the Conflicts Extractor (part 1 of 3).
11	Code of Ethics, Personal Trading, Participation in Client Transactions	Personal-trading policy, material conflicts. Read by the Conflicts Extractor (part 2 of 3).
12	Brokerage Practices	Soft dollars, directed brokerage, trade aggregation. Read by the Conflicts Extractor (part 3 of 3).
13	Review of Accounts	Account-review cadence and triggers.
14	Client Referrals & Other Compensation	Referral arrangements.
15	Custody	Custody arrangements and account-statement delivery.
16	Investment Discretion	Discretionary authority granted.
17	Voting Client Securities	Proxy-voting policy.
18	Financial Information	Pre-paid fees, financial condition, bankruptcy disclosures.

ADV-Lens reads **Items 5, 9, 10, 11, 12** with dedicated LLM extractors. The other 13 Items are segmented and stored but not currently scored — they're available for future extractors.

3.1 What the input actually looks like

The three images below are from the live Brown Advisory run on 2026-04-26 — a deliberately neutral mid-size RIA, no recent compliance noise, multi-program brochure structure (the kind that exercises both the regex segmenter and the LLM rescue per ADR 0014).

The IAPD firm summary page — what a CRD resolves to. The "PART 2 BROCHURES" button links to the per-brochure PDF the system actually fetches. Brown Advisory: CRD 110181, SEC# 801-38826, registered 6/7/1991.

 Investment Adviser Public Disclosure

INDIVIDUAL



FIRM

Individual Name/CRD#

Firm Name or CRD/SEC# (optional)

City, State or ZIP (optional)

at

in



When communicating online or investing with any professional, make sure you know who you're dealing with. [Imposters](#) might link to sites like BrokerCheck websites, or through [social media](#), trying to steal your personal information or your money. Please contact FINRA with any concerns.

[< Back to Results](#)

Investment Adviser Firm Summary

BROWN ADVISORY (CRD # 110181/SEC#:801-38826)

BEATY HAYNES & ASSOCIATES, INC., BROWN ADVISORY LLC, BROWN ADVISORY, BEATY HAYNES LLC, BEATY HAYNES & PATTERSON INC

[VIEW LATEST FORM ADV FILED](#)[PART 2 BROCHURES](#)[PART 3 RELATIONSHIP SUMMARY](#)The adviser's **REGISTRATION** status is listed below.

REGISTRATION STATUS

SEC / JURISDICTION	REGISTRATION STATUS 	
SEC	Approved	6/7/1991

NOTICE FILINGS

Investment adviser firms registered with the SEC may be required to provide to state securities authorities a copy of their Form ADV and any filed with the SEC. These filings are called "notice filings". Below are the states with which the firm you selected makes its notice filings. Also became notice filed or registered in each state.

JURISDICTION	EFFECTIVE DATE
Alabama	3/16/2012
Arizona	3/16/2012
California	3/27/2003
Colorado	3/16/2012
Connecticut	3/16/2012
Delaware	3/16/2012
District of Columbia	1/15/1994
Florida	9/25/1996
Georgia	3/16/2012
Hawaii	3/28/2024
Idaho	3/31/2017
Illinois	3/19/2012
Indiana	3/29/2019
Kentucky	3/31/2017
Louisiana	3/16/2012
Maine	3/16/2012

Maryland	10/3/1991
Massachusetts	3/31/2011
Michigan	3/16/2012
Minnesota	3/16/2012
Missouri	3/16/2012

Item 5 — Fees and Compensation (page 13 of the brochure). A multi-program tiered AUM fee schedule. Each program lists breakpoints in plain English; the fee extractor's job is to recover these into structured `FeeSchedule` / `FeeTier` objects.

0.25% on the next \$250 million under management
0.20% on amounts over \$500 million under management

LARGE-CAP SUSTAINABLE GROWTH (ACCOUNTS BELOW \$150M)

0.80% on the first \$10 million under management
0.60% on the next \$15 million under management
0.50% on the next \$25 million under management
0.40% on amounts over \$50 million under management

LARGE-CAP SUSTAINABLE GROWTH (ACCOUNTS ABOVE \$150M)

0.46% on the first \$150 million under management
0.43% on the next \$100 million under management
0.40% on the next \$250 million under management
0.35% on amounts over \$500 million under management

FLEXIBLE EQUITY (ACCOUNTS BELOW \$150 MILLION)

0.60% on the first \$25 million under management
0.50% on the next \$25 million under management
0.45% on the next \$50 million under management
0.35% on the next \$50 million under management

FLEXIBLE EQUITY (ACCOUNTS ABOVE \$150 MILLION)

0.45% on the first \$150 million under management
0.275% on the next \$100 million under management
0.25% on the next \$250 million under management
0.20% on amounts over \$500 million under management

MID-CAP GROWTH (ACCOUNTS BELOW \$150 MILLION)

0.75% on the first \$50 million under management
0.50% on the next \$50 million under management
0.475% on the next \$50 million under management

MID-CAP GROWTH (ACCOUNTS ABOVE \$150 MILLION)

0.58% on the first \$150 million under management
0.45% on the next \$100 million under management
0.425% on the next \$250 million under management
0.35% on amounts over \$500 million under management

SMALL-CAP GROWTH, SMALL-CAP FUNDAMENTAL VALUE, SUSTAINABLE SMALL-CAP CORE

1.00% on the first \$25 million under management
0.90% on the next \$25 million under management
0.80% on the next \$50 million under management

0.70% on amounts over \$100 million under management

Item 9 — Disciplinary Information (page 43 of the brochure). The canonical "no disciplinary history" disclosure pattern: "*Neither Brown Advisory nor any of our supervised persons have been involved in any legal or disciplinary events...*". The disciplinary extractor's job is to recognise this as `has_disciplinary_history: false` rather than hallucinating events from the surrounding text.

performance calculations for clients who hold alternative and difficult-to-value assets. Investments in alternative investments will be inaccurate to the extent they rely on valuations that are not current and accurate.

ITEM 9 DISCIPLINARY INFORMATION

Neither Brown Advisory nor any of our supervised persons have been involved in any disciplinary events (i.e., criminal or civil action in a domestic, foreign or military, or administrative proceeding before the SEC, any other federal regulatory agency, any state regulatory agency or self-regulatory organization) that are material to evaluating our business or the integrity of our management.

ITEM 10 OTHER FINANCIAL INDUSTRY ACTIVITIES AND AFFILIATIONS

BAGH, a Delaware limited liability company, serves as the parent company of BAI and BAM. BAM serves as the manager of BAGH and the managing member of BAM. BAM is a holding company that serves as the parent company to several Brown Advisory subsidiaries.

Brown Advisory is a registered investment adviser with the SEC and is a wholly owned subsidiary of BAM. Brown Advisory is eligible to conduct certain activities with permitted types of non-registered and registerable activities in Ontario, Quebec, Nova Scotia, and New Brunswick in reliance on the International Adviser Exemption.

FUTURES COMMISSION MERCHANT, COMMODITY POOL OPERATOR, COMMODITY TRADING REGISTRATION STATUS

Brown Advisory is not registered with the U.S. Commodity Futures Trading Commission ("CFTC") as a Commodity Trading Advisor ("CTA") or as a Commodity Pool Operator ("CPO"). Certain employees and members of management serve as "associated persons" of affiliates of Brown Advisory that are registered with the CFTC as a CTA or CPO or are exempt from registration.

RELATED PERSONS

Brown Advisory has certain relationships or arrangements with related persons that are material to its advisory business or its clients. Below is a description of such relationships and some potential conflicts of interest that arise from them. Brown Advisory has adopted policies and procedures reasonably designed to prevent, limit or mitigate conflicts of interest that may arise from such relationships between Brown Advisory and its affiliates.

AFFILIATIONS WITH INVESTMENT COMPANIES OR OTHER POOLED INVESTMENT VEHICLES

Brown Advisory has arrangements that are material to its advisory business with various types of investment companies. Brown Advisory serves as the investment adviser to affiliated funds, ETFs, CITs, investment trusts, and UCITS funds. We may also serve as the managing member of private funds that invest in public and private securities.

Brown Advisory has arrangements to serve as adviser or sub-adviser to investment companies and pooled investment vehicles sponsored by other unaffiliated financial services firms.

4. What comes out: the Redline Report

The system's output is a `RedlineReport` Pydantic object, serialised to JSON and (after the audit-trail bundle work — see § 9) optionally rendered as a 1-page HTML/PDF for printing or emailing to the CCO.

4.1 Structure

A `RedlineReport` contains:

- `brochure_crd` and `brochure_version_id` — exact provenance.
- `scorecard` — an overall score (0-100) plus four mandatory category sub-scores, each 0-100 with a one-sentence rationale:
 - **compliance** — does the brochure meet SEC plain-English and coverage expectations?
 - **transparency** — are fees, conflicts, and risks disclosed clearly?
 - **conflicts_handling** — when conflicts exist, is the mitigation explained?
 - **fee_competitiveness** — relative to the AUM-band peer cohort.
- `findings` — between 4 and 12 individual flagged items, each with:
 - a stable ID (`F-001`, `F-002`, ...),
 - a `category` tag (e.g. `compliance_program`, `fee_structure`, `conflicts`),
 - a severity (`info` / `low` / `medium` / `high` / `critical`),
 - an `item_reference` — the ADV Item number the finding cites (null for cross-Item issues),
 - a `summary` (one sentence) and a `detail` (one paragraph),
 - a `sec_expectation_ref` pointing at the specific Form ADV instruction or Advisers Act rule the expectation derives from,
 - an optional `peer_comparison` populated when peer context was available,
 - a `recommendation` — what the CCO should actually do.
- `peer_comparisons` — list of cross-firm comparisons keyed on extracted field, populated when peer context was available.
- `extraction_warnings_seen` — surfaces any `extraction_warnings` that the upstream extractors emitted, so the redline writer's interpretive judgment is auditable.
- `notes` — free-form notes from the redline writer about caveats, upstream input quality, or limits on the assertions made.

The schema lives in `src/adv_lens/extractors/schemas.py` (`RedlineReport`, `Scorecard`, `ScoreCategory`, `Finding`).

4.2 Sample (real run, 2026-04-26 — with LLM rescue)

The first live end-to-end run against a real IAPD brochure: **Brown Advisory LLC** (CRD 110181, BRCHR_VRSN_ID 1037550, brochure filed 2026-03-31). Full artifact at [docs/examples/sample-report.json](#) (19KB, redline + provenance metadata) and the full pipeline state at `docs/examples/sample-state.json` (~290KB).

Honest commentary on this sample. The overall_score is 68. The brochure has a non-canonical multi-program structure — only Items 2/3/4/9/15/16 have ALL-CAPS narrative headers; Items 5/10/11/12 lack standalone section headers because their content is bundled into per-program subsections. The pipeline handled this in two stages:

1. The regex `HeuristicSegmenter` caught Items 2/3/4/9/15/16 cleanly (Item 9 = no disciplinary disclosures, a score-positive signal) but produced 0–144-char fragments for Items 5/10/11/12.
2. The **ADR 0014 LLM rescue** kicked in: Haiku 4.5 located the real spans for the four missing Items in one prompt round-trip, producing 3,452 / 3,813 / 16,544 / 18,120-char bodies for Items 10 / 11 / 5 / 12 respectively. Backend tag on the resulting `SegmentedBrochure` is `heuristic+llm_fallback`, with a warning recording exactly which Items were rescued.

Qdrant remained offline for this run, so `retrieve_peers` degraded gracefully (empty `peer_context`) and the redline writer noted the absence of peer benchmarking in its headline.

Of the 11 findings, 3 are high-severity (re-extraction of misaligned Items 11/12 spans recommended), 4 medium, 2 low, 2 info. F-001 is a

positive

info-severity finding documenting that the Private Client

\$5M schedule's 100→30 bp tiered structure is industry-standard.

The portfolio claim is now demonstrable end-to-end: "we extract Items 5/9/10/11/12 from canonical and multi-program brochures alike, with a deterministic-first / LLM-rescue-as-fallback architecture, and surface any remaining gaps as findings rather than hiding them." Full diagnostic in [ADR 0014](#); the SEC URL/UA migration story patched in this same session is in [ADR 0015](#).

```
{
  "meta": {
    "crd": "110181",
    "brochure_version_id": "1037550",
    "trace_id": "brown-advisory-rescued",
    "review_status": "pending_review",
    "segmenter_backend": "heuristic+llm_fallback"
  },
  "redline": {
    "scorecard": {
      "overall_score": 68,
      "headline": "Brown Advisory (CRD 110181) brochure presents a clean disciplinary record and a competitive tiered fee structure, but the extracted text for Items 11 and 12 was misaligned and key disclosures (Code of Ethics, brokerage practices, performance fees, wrap economics) could not be verified from this extraction; scorecard generated without peer comparison and several findings are gated on a re-extraction of the source brochure.",
      "categories": [
        {
          "name": "compliance",
          "score": 65,
          "rationale": "No disciplinary history disclosed (positive). Extracted text for Items 11 (Code of Ethics) and 12 (brokerage) was misaligned with their headers, leaving core compliance-program controls unverifiable from this extraction."
        },
        {
          "name": "transparency",
          "score": 60,
          "rationale": "Private Client >$5M schedule is well-tiered and clearly stated. Other programs are weaker: <$5M schedule does not state rates above $5M; E&F and OCIO are disclosed only as maximum rates with no breakpoints; wrap-program compensation to the firm is not quantified."
        },
        {
          "name": "conflicts_handling",
          "score": 65,
          "rationale": "Standard but material proprietary-product conflict surfaced (firm may recommend affiliated funds/CITs in which it has a financial interest). No B/D, insurance, or banking affiliation footprint visible in the extraction."
        },
        {
          "name": "fee_competitiveness",
          "score": 78,
          "rationale": "Headline rates are reasonable: Private Client reaches 30 bps at $100M+, OCIO is capped at 30 bps, E&F is capped at 75 bps. Fees stated to be negotiable. Without peer context this is an absolute, not relative, assessment."
        }
      ]
    },
    "findings": [
      {
        "id": "F-001",
        "category": "fee_structure",
        "severity": "info",
        "item_reference": 5,
        "summary": "Tiered AUM fee schedule for Private Client Portfolios >$5M is competitive and uses standard breakpoints.",
        "detail": "The Private Client Portfolios >$5M schedule begins at 100 bps on the first $5M and steps down through 75/50/35/30 bps, with a top breakpoint at $100M. Billing is quarterly and fees are stated to be negotiable. Structure appears consistent with industry norms for high-net-worth discretionary management.",
        "sec_expectation_ref": "Form ADV Part 2A Instructions, Item 5.A (fee schedule disclosure)",
        "peer_comparison": null,
        "recommendation": "No action required. Document the schedule in the firm's fee-comparison file for next annual review."
      },
      {
        "id": "F-002",
        "category": "fee_structure",
        "severity": "low",
        "item_reference": 5,
        "summary": "Private Client <$5M schedule is ambiguous: only the first $5M is tiered; no rate stated for assets above $5M.",
        "detail": "The <$5M schedule shows 125 bps on the first $3M and 100 bps from $3M-$5M, but does not state a rate for assets above $5M. The extractor's working hypothesis is that this schedule applies only when the $5M minimum is waived. The brochure language as extracted may leave a sub-$5M client unclear about how growth above $5M would be billed.",
        "sec_expectation_ref": "Form ADV Part 2A Instructions, Item 5.A (plain-English fee description)",
        "peer_comparison": null,
      }
    ]
  }
}
```

```
      "recommendation": "Review the source brochure text to confirm whether the schedule rolls into the >$5M grid once a sub-minimum account crosses $5M. If ambiguous, escalate to disclosure counsel for a clarifying edit at the next amendment."
    }
  ]
}
```

The full report has 11 findings (3 high, 4 medium, 2 low, 2 info). The 3 high-severity findings recommend re-extracting Items 11 and 12 where the LLM rescue produced misaligned spans the redline writer detected in its content review. Read [docs/examples/sample-report.json](#) for the complete artifact including provenance (`brochure_sha256` , `report_hash`) used by the audit-trail mechanism.

4.3 How to read the score

- **90+** — exemplary. Use as a peer-benchmark reference.
- **80-89** — solid. Address 2-3 medium findings before next annual amendment; no exam concern.
- **70-79** — adequate. Address all medium and high findings before next amendment; prepare a remediation memo for exam file.
- **60-69** — weak. Triggers a formal disclosure-improvement project.
- **Below 60** — material disclosure gap. Engage outside compliance counsel; do not rely on the brochure as currently filed.

These thresholds are advisory, not regulatory. Your compliance counsel sets the actual remediation bar.

4.4 How to read the findings

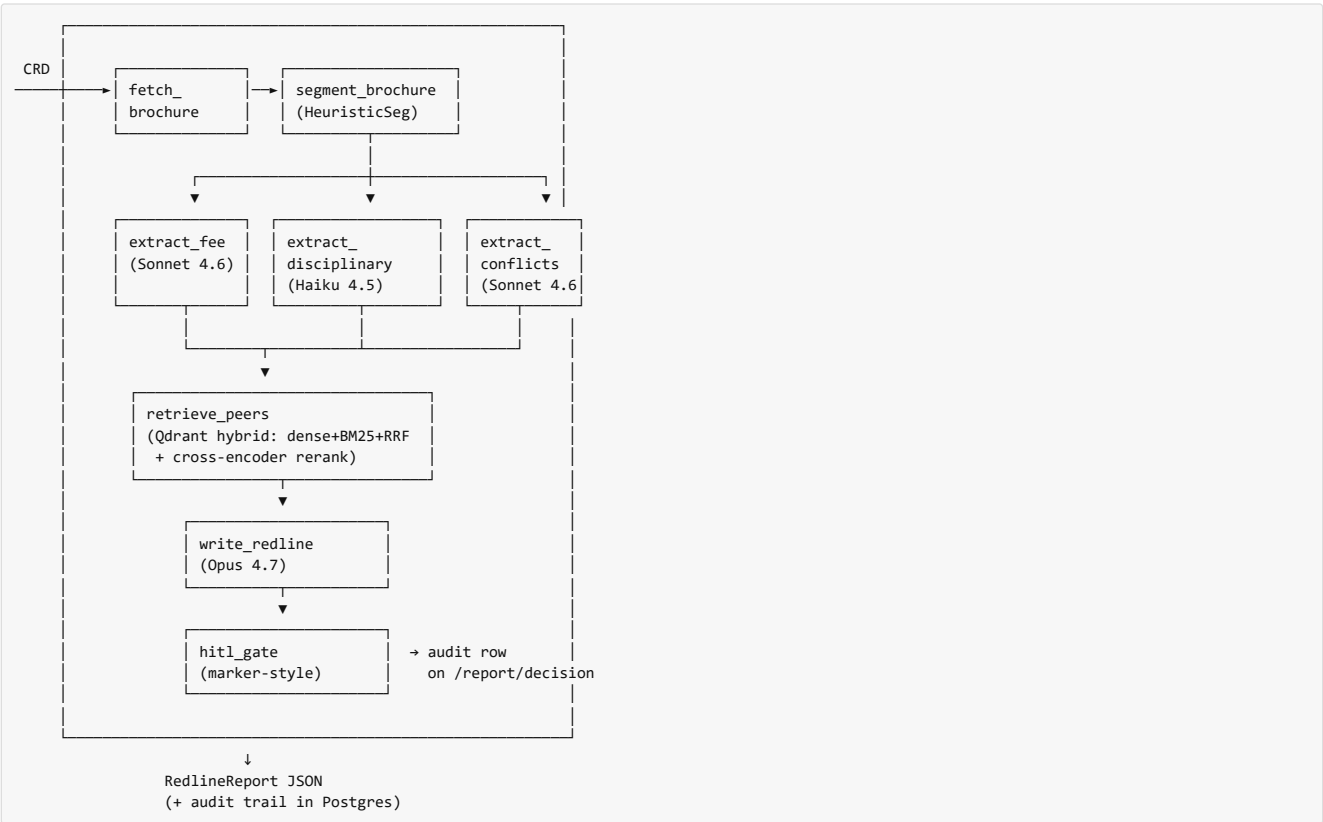
Each finding cites the ADV Item it relates to and has a severity. The severity scale is calibrated to **what would change at exam**, not to what would change a hiring decision:

- **critical** — likely violates a specific Advisers Act rule or Marketing Rule provision. Fix before any new client onboarding.
- **high** — material disclosure gap an SEC examiner would request a written explanation for. Fix before next amendment.
- **medium** — clarity or completeness gap that would draw a comment letter but not a deficiency finding. Fix in next amendment cycle.
- **low** — stylistic or minor structural improvement. Address opportunistically.
- **info** — observation, not a defect. Common when the system finds something noteworthy but not actionable.

A finding without a numeric Item reference signals a cross-Item issue (e.g., contradiction between Item 5 fees and Item 12 brokerage).

5. How the system works

5.1 Pipeline at a glance



A live Mermaid version of the same diagram is in `docs/architecture.md` (§ Pipeline) — GitHub renders it inline.

5.2 ETL at each hop

For each arrow in the diagram, this table names the **Extract**, the **Transform**, and the **Load** so a CCO can trace any field in the final scorecard back to a source artifact, a transform, and an audit row.

Hop	Extract	Transform	Load
CRD → IAPD search	HTTPS GET to <code>api.adviserinfo.sec.gov</code> (rate-limited 5 rps)	JSON → BrochureRef Pydantic	<code>data/brochures/<CRD>/index.json</code>
IAPD search → PDF	HTTPS GET to <code>files.adviserinfo.sec.gov/...pdf</code>	bytes → SHA-256 + cache key	<code>data/brochures/<CRD>/<BRCHR_VRSN_ID>.pdf</code> (idempotent)
PDF → text	<code>pypdf</code> page iteration	text concat with page boundaries	in-memory <code>str</code> (passed to segmenter)
text → 18 sections	HeuristicSegmenter (regex on Item 1 / Item 2 / ... headings)	<code>dict[ItemNumber, SectionBody]</code> → SegmentedBrochure Pydantic	in-memory; persisted in <code>state.segmented_brochure</code>
Item 5 → fee extraction	LLM call (Sonnet 4.6 via Instructor)	structured JSON → FeeExtraction Pydantic	one row in <code>llm_calls</code> (input/output/tokens/cost), one field in <code>state.extractions</code>
Item 9 → disciplinary extraction	LLM call (Haiku 4.5 via Instructor)	structured JSON → DisciplinaryExtraction Pydantic	one row in <code>llm_calls</code> , one field in <code>state.extractions</code>
Items 10/11/12 → conflicts extraction	LLM call (Sonnet 4.6 via Instructor)	structured JSON → ConflictsExtraction Pydantic	one row in <code>llm_calls</code> , one field in <code>state.extractions</code>
extractions → peer hits	Qdrant query per Item (5/9/10/11/12) with AUM-band filter, exclude self-CRD	dense <code>bge-small-en-v1.5</code> + sparse hashed BM25 → RRF fusion → cross-encoder rerank (<code>ms-marco-MiniLM-L-6-v2</code>)	<code>state.peer_context: list[dict]</code> (PeerHit dumps)
extractions + peers → redline	LLM call (Opus 4.7 via Instructor)	structured JSON → RedlineReport Pydantic + structural validator	one row in <code>llm_calls</code> , <code>state.redline</code>

Hop	Extract	Transform	Load
redline → review marker	compute_report_hash(redline) SHA-256	sets state.review_status="pending_review", state.report_hash	pipeline_runs.result (full state JSON)
CCO POST /report/decision	HTTP request body	HumanReviewDecision Pydantic	one row in human_reviews (decision, reviewer, rationale, report_hash)

Every LLM call additionally emits a Langfuse trace span when Langfuse is configured (one trace per `trace_id`, one span per node).

5.3 Per-node breakdown

See docs/architecture.md § Pipeline (current code) for the full, code-citing description of each node. The short version:

- **fetch_brochure** resolves the most recent brochure version for the CRD (or uses a caller-supplied `BRCHR_VRSN_ID`), fetches the PDF with a content-addressed cache, populates SHA-256 + cache-status flags on the state.
- **segment_brochure** runs the regex segmenter; missing Items land in `state.errors` rather than raising. This means a brochure missing Item 9 still flows through the rest of the pipeline.
- **extract_fee / extract_disciplinary / extract_conflicts** run in **parallel** off the segmenter via LangGraph's parallel-edges pattern with a custom `merge_extractions` reducer (ADR 0006). Each returns only its own field; the reducer composes them.
- **retrieve_peers** is the fan-in node. It waits for all three extractors to complete before it issues per-Item Qdrant queries.
- **write_redline** is the single Opus 4.7 call per pipeline run. Output is structurally validated before it leaves the node.
- **hitl_gate** is **marker-style** (sets `review_status="pending_review"`
 - `report_hash`), not a true LangGraph `interrupt_before`. ADR 0010 documents why.

5.4 Per-node model assignment

Node	Model	Why
fetch_brochure	(no LLM)	Pure I/O
segment_brochure	(no LLM)	Regex
extract_fee	Sonnet 4.6	Numeric tables + categorical fields; benefits from Sonnet's structured-output discipline
extract_disciplinary	Haiku 4.5	Disciplinary disclosures are short and largely categorical; Haiku is fast and cheap
extract_conflicts	Sonnet 4.6	Three concatenated Items, lots of conditional fields; Sonnet handles the longer context
retrieve_peers	(no LLM)	Vector + lexical search; cross-encoder rerank is a bge model
write_redline	Opus 4.7	Synthesises 4 inputs into a defensible scorecard + findings; one call per run is the cost cap

Per-node assignments are environment-overridable via `MODEL_*` env vars (see § 10.3).

6. Running the system

6.1 Prerequisites

- Python 3.12 + `uv` (see `pyproject.toml` for exact deps)
- Docker + Docker Compose (for Postgres + Qdrant + Langfuse)
- `ANTHROPIC_API_KEY` in `.env` for any extraction or redline run
- Optional: `LANGFUSE_*` keys after first compose-up, for trace visibility

6.2 First boot

```
cp .env.example .env
# edit .env: paste ANTHROPIC_API_KEY, leave LANGFUSE_* blank for now
docker compose up -d
# visit http://localhost:3000 to provision Langfuse, paste back the
# LANGFUSE_PUBLIC_KEY / LANGFUSE_SECRET_KEY
uv run python -m adv_lens.app.main # or: docker compose up app
```

Health check: `GET http://localhost:8000/healthz`

6.3 One-shot CLI (synchronous; useful for evals and debugging)

```
# Fetch + segment only (no LLM cost)
uv run python -m adv_lens.app.graph.cli 108000 --no-llm

# Full pipeline (extractors + redline + HITL marker)
uv run python -m adv_lens.app.graph.cli 108000

# Pin a specific brochure version
uv run python -m adv_lens.app.graph.cli 108000 --brochure-version 987654
```

CLI prints the final `ADVState` as JSON. Pipe to `jq` for inspection.

6.4 HTTP (asynchronous; production shape)

```
# Submit
curl -X POST http://localhost:8000/pipeline/run \
  -H 'content-type: application/json' \
  -d '{"crd": "108000"}'
# → 202 {"trace_id": "advLens-abc123", "status": "queued",
#       "status_url": "/pipeline/run/advLens-abc123"}

# Poll
curl http://localhost:8000/pipeline/run/advLens-abc123
# → 200 {"trace_id": ..., "status": "running" | "complete" | "failed",
#       "result": {... full ADVState when complete ...}}
```

The status walks `queued` → `running` → `complete` | `failed`. Polling cadence: 5-15 seconds is plenty; a typical end-to-end run is 30-90 s on a real brochure.

6.5 Approving or rejecting a report (HITL)

When `result.review_status == "pending_review"`, the CCO acts:

```

curl -X POST http://localhost:8000/report/decision \
-H 'content-type: application/json' \
-d '{
  "trace_id": "advlens-abc123",
  "report_hash": "<echoed from the pipeline result>",
  "brochure_crd": "108000",
  "decision": "approved",
  "reviewer": "jane.cco@firm.example",
  "rationale": "Reviewed all medium findings; concur with the action items in F-001 and F-002."
}'
# → 201 + DecisionResponse including the audit row id

# Read the decision history for a run:
curl http://localhost:8000/report/decision/advlens-abc123
# → list of decisions oldest-first; re-posts deliberately create new rows

```

Decision options: `approved`, `rejected`, `revise_requested`.

6.6 The audit trail

Three Postgres tables, queryable at any time:

- `llm_calls` — one row per LLM invocation. Includes input, output, model ID, intended temperature, token counts, USD cost, trace ID, node name, brochure CRD, timestamp.
- `human_reviews` — one row per CCO decision. Includes decision, reviewer, rationale, `report_hash`, timestamp.
- `pipeline_runs` — one row per pipeline invocation. Includes status, started/completed timestamps, full final state JSON, error.

A typical audit query: "show me every LLM call for trace X with model and cost":

```

SELECT node, model, prompt_tokens, completion_tokens, cost_usd, created_at
FROM llm_calls WHERE trace_id = 'advlens-abc123' ORDER BY created_at;

```

NOT BUILT YET — single-command audit-trail bundle.

Today the audit lives across three tables and a JSON column. A `GET /report/audit/{trace_id}` endpoint that returns a single ZIP containing the cached input PDF, the per-Item extraction JSONs, the RedlineReport, all `llm_calls` rows, and all `human_reviews` rows would close the "give me the exam-defensible bundle" need in one request. **Target plan:** Week-5 polish; Jinja+ZIP, ~3 hours.

7. Safety + compliance

7.1 What this is, what it isn't

Is: an analyst aid for CCOs and M&A diligence teams reading public SEC filings. Structured extraction + peer benchmarking + redline.

Isn't:

- legal advice,
- an auto-publisher of redacted or revised brochures,
- a recommendation engine for investment products,
- a substitute for CCO judgment.

Every report passes through a human review gate before it leaves the system.

7.2 The HITL gate (built)

The `hitl_gate` node runs at the end of every pipeline. It does **not** publish the report. It marks the report as `pending_review` and computes a `report_hash` (SHA-256 of the canonical RedlineReport JSON). The CCO acts via `POST /report/decision`, which writes a row to `human_reviews` keyed on the hash.

A re-run of the pipeline against the same CRD produces a new `trace_id` but a `report_hash` that may or may not match the prior approval. This is intentional — approvals are pinned to the exact bytes the CCO read, not to a moving target.

7.3 Audit-table coverage (built)

Every LLM call and every human decision lands in Postgres with a trace ID that joins back to the pipeline run. The audit posture is designed to answer:

"Show me how this finding was produced and who signed off on it."

See § 6.6 for query examples.

7.4 Regulatory posture

ADV-Lens reads exclusively public SEC filings. No client PII, no gated scraping. The companion `docs/compliance.md` enumerates the specific SEC and FINRA rules the system engages with.

NOT BUILT YET — full compliance doc.

`docs/compliance.md` is currently a placeholder. **Target plan:** Week-5 polish pass covers: SEC Marketing Rule 206(4)-1 implications (none, but defensible), Advisers Act recordkeeping rule 204-2, the Form ADV instructions ADV-Lens uses as a scoring rubric, and the exam-defensibility argument for the audit posture. Co-lands with the README final pass and this manual's revision.

7.5 Data handling

- **Input PDFs** cached on disk under `data/brochures/<CRD>/<BRCHR_VRSN_ID>.pdf`. Public SEC filings; no expiry policy needed.
 - **Extractions and redlines** persisted in `pipeline_runs.result`. 90-day retention is the planned default; today operators clean up manually.
 - **LLM call logs** in `llm_calls` retain everything indefinitely for audit. Cost is tracked per row.
 - **No client PII ever transits the system.** ADV-Lens has no notion of client accounts and refuses (by design) to accept anything other than a CRD + optional version ID as input.
-

8. Production checklist

8.1 Daily / per-shift

- `GET /healthz` returns 200.
- Any `pipeline_runs` row stuck in `running` longer than 10 minutes gets reaped by the stuck-row reaper (see § 8.3).
- Langfuse dashboard shows per-pipeline trace health.

8.1.1 Wiring Langfuse to your runs

Langfuse trace emission is plumbed and tested in code: every LLM call in `LLMClient.extract` emits one Langfuse generation span, and all spans for one pipeline invocation roll up under a deterministic trace ID (`Langfuse.create_trace_id(seed=trace_id)`). The wire is no-op when Langfuse isn't configured. To turn it on:

1. **Bring up the self-host** (already in `docker-compose.yml`):

```
docker compose up -d langfuse-web langfuse-worker langfuse-db
```

2. **Provision keys** in the Langfuse UI at <http://localhost:3000>: create an organisation + project, copy the public/secret keys.
3. **Paste keys into** `.env`:

```
LANGFUSE_PUBLIC_KEY=pk-1f-...
LANGFUSE_SECRET_KEY=sk-1f-...
LANGFUSE_HOST=http://localhost:3000
```

4. **Run the pipeline as usual**. The next CLI or HTTP invocation emits traces. Trace URL for any `trace_id` is available via `adv_lens.app.observability.get_trace_url(trace_id)` and printed in `logs/` when configured.

Coverage today: one Langfuse generation span per LLM call (fee / disciplinary / conflicts extractors, segmenter LLM rescue, redline writer), each with model ID, input/output tokens, USD cost, the brochure CRD, and the originating pipeline trace ID. Roll-up trace contains every span for one pipeline invocation in time order.

README trace links populate after the first traced run. The README's "3 example traces" placeholders are populated by hand from the Langfuse UI after the self-host has at least one CCO-approved brochure run end-to-end.

8.2 Weekly

- Run the eval harness against the golden set:

```
uv run python -m eval.runner
```

Compare F1 per section to the prior week's baseline (committed to `eval/results/`). A drop of more than 0.05 mean F1 in any section warrants a prompt or schema diff.

8.3 The stuck-row reaper

Process restarts (deploys, OOMs, network partitions) leave any in-flight pipeline runs stuck in `status="running"`. The reaper sweeps them:

```
# Dry run - show candidates without mutating
uv run python -m adv_lens.app.jobs.reaper --dry-run --verbose

# Actually mark stuck rows as failed
uv run python -m adv_lens.app.jobs.reaper --verbose
```

Wire into cron / a Kubernetes CronJob at whatever cadence matches your deployment's restart frequency. Default threshold is 10 minutes; tune via `PIPELINE_RUN_REAP_THRESHOLD_MINUTES` in `.env`.

8.4 Anthropic spend

Every LLM call writes its cost estimate in USD to `llm_calls.cost_usd`. A typical full pipeline run costs 0.50–1.50 (one Opus call dominates). Per-day budget alarm:

```
SELECT DATE(created_at) AS day, SUM(cost_usd) AS daily_usd
FROM llm_calls
WHERE created_at > NOW() - INTERVAL '7 days'
GROUP BY day ORDER BY day;
```

8.5 Eval-harness regression policy

NOT BUILT YET — CI eval-regression gate.

Today the eval harness runs locally and writes `eval/results/<UTC>/report.{json,md}`. No CI check enforces a regression bar yet. **Target plan:** Week-4 — GitHub Actions job that runs the eval on every PR (against an Anthropic-key secret), compares the resulting `mean_score` per section to the latest committed baseline, and blocks the PR if any section drops more than 0.05. Pairs with the LLM-as-judge dual-judge pattern (ADR 0009).

9. What's not built yet — roadmap

The system is feature-complete on the **pipeline + extractor + audit** axes (Weeks 1-3). The remaining work is **rigour + presentation + operational polish** (Weeks 4-6). In rough priority order:

9.1 Week 4 — eval rigour + segmenter robustness

- **Segmenter robustness against multi-program / bundled-items brochures** (see ADR 0014). The `HeuristicSegmenter` regex looks for line-anchored "Item N" headers (case-insensitive, common punctuation variants). It works cleanly on the textbook brochure structure but breaks on brochures that **bundle items together** — for example, where the fee-disclosure narrative for Items 5/10/11/12 lives inside per-program subsections like "Brown Advisory Wrap Program — Fees and Compensation" rather than a standalone "Item 5" header. The Brown Advisory live run (§ 4.2) surfaced this honestly: Items 2/3/4/9/15/16 parsed cleanly, Items 5/10/11/12 did not. **Target plan:** add an LLM fallback (Haiku 4.5, one call per brochure) that locates item-section spans when the regex finds <2K-char bodies for any of Items 5/9/10/11/12. Triggered only on the difficult subset; regex stays primary. Full design + diagnostic in ADR 0014.
- **LLM-as-judge for redline scoring** (ADR 0009). Today the redline is scored only structurally (4-12 findings, valid scorecard categories, unique IDs, severity not pathological). A second LLM judge reading the redline against the brochure would give a content-quality score. Plan: dual-judge with cross-checking to control for judge drift.
- **Golden set scaling: 19 → 65 fixtures**. Spread across all three extractor sections; mix synthetic-clean and realism-style prose. The 19 fixtures shipped today validate the extractors structurally; 65 gives stable per-section F1 numbers that survive run-to-run noise.
- **CI eval-regression gate** (see § 8.5).
- **Multi-run averaging**. Without `temperature=0` (deprecated by Anthropic on Opus 4.7, with Sonnet/Haiku to follow), single-shot F1 has run-to-run variance up to 0.15 on individual fixtures. Eval should run each fixture N=3 and report median + spread.

9.2 Week 5 — presentation polish

- **Per-brochure HTML/PDF redline report: Done 2026-04-26**. `adv_lens.redline.cli` renders any `RedlineReport` JSON (or the trimmed sample-report wrapper) to a self-contained HTML artifact and optional PDF via headless Chrome. Score gauge, severity-coloured finding cards, category breakdown table, footer with provenance (CRD, version, brochure SHA-256, trace, report hash). Live sample at docs/examples/sample-report.html

ADV-LENS COMPLIANCE REDLINE

CRD 110181 — Brochure 1037550

Form ADV Part 2A scorecard, generated 2026-04-26 21:18 UTC

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OVERALL SCORE

HEADLINE

Brown Advisory (CRD 110181) brochure presents a clean disciplinary record and a competitive tiered fee structure, but the extracted text for Items 11 and 12 was misaligned and key disclosures (Code of Ethics, brokerage practices, performance fees, wrap economics) could not be verified from this extraction; scorecard generated without peer comparison and several findings are gated on a re-extraction of the source brochure.

Category breakdown

Category	Score	Rationale
Compliance	65	No disciplinary history is disclosed (positive). However, the extracted text for Items 11 (Code of Ethics / personal trading) and 12 (brokerage practices) was misaligned with their headers, leaving core compliance-program controls unverifiable from this extraction. Rule 204A-1 obligations and Item 12 soft-dollar/directed-brokerage posture must be confirmed before the CCO signs off. Score reflects a clean disciplinary record offset by audit gaps that are extraction-driven rather than substantive.
Transparency	60	The Private Client >\$5M schedule is well-tiered and clearly stated. Several other programs are weaker: the <\$5M schedule does not state rates above \$5M; E&F and OCIO are disclosed only as maximum rates with no breakpoints; wrap-program compensation to the firm is not quantified in the extracted text; and model-delivery fee terms are absent. Performance-fee posture (Item 6) is also missing from the extraction.
Conflicts Handling	65	The extraction surfaces a standard but material proprietary-product conflict: the firm may recommend affiliated mutual funds, ETFs, CITs, and private funds in which it has a material financial interest. The affiliation footprint as extracted is relatively contained (no broker-dealer, insurance, or bank affiliate indicated). However, neither the Code of Ethics framework nor the Item 12 brokerage practices were captured, so the firm's mitigation posture for these conflicts cannot be scored from this extraction alone.
Fee Competitiveness	78	Headline rates are reasonable for the segments served: the Private Client schedule reaches 30 bps at \$100M+, OCIO is capped at 30 bps, and E&F is capped at 75 bps. Fees are stated to be negotiable on the principal schedules. Without peer context this is an absolute, not relative, assessment.

Findings (11)

- **Langfuse traces wired** (see § 8.1).
- **Compliance doc final pass** (see § 7.4).
- **Demo GIF / video**. 60-90 seconds, full pipeline against one brochure ending with a CCO approval action. README embed.
- **Visual sample of input data** in this manual (see § 3 callout).
- **Live IAPD run** — produces the real `docs/examples/sample-report.json` that updates § 4.2 in this manual.
- **Audit-trail bundle export** (see § 6.6 callout).

- **Output bundle layout** — design once for the CLI and HTTP endpoint: per-run output directory containing the cached PDF, the segmented sections, the extraction JSONs, the RedlineReport, and the audit excerpt as a single bundle.

9.3 Week 6 — optional bolt-on

- **ADV-Diff** — quarterly change detector. When a watched CRD's `BRCHR_VRSN_ID` changes on IAPD, fire the pipeline against the new brochure, diff the resulting RedlineReport against the prior approved one, and route the change summary to the CCO. Reuses every component of this system.

9.4 Post-portfolio

- **Peer benchmarking dashboard** — for a single brochure, show its scores vs the AUM-band median + p25/p75 per category. The actual product story for an F2-Strategy-style consulting pitch. Needs $N \geq 10$ real brochures cached, so a Week-5+ thing.
 - **Eval-trend dashboard** — static HTML page from `eval/results/*.json`, F1 per section over commit history. GitHub Pages publish.
 - **On-prem branch** — Ollama (Qwen 2.5 / Llama 3.3) fallback for customers who can't ship brochure text to a hosted LLM. ADR 0012 pending.
 - **Peer auto-discovery from IARD** — today the peer corpus is operator-curated. Auto-discovery from the IARD bulk Part 1 CSV by AUM band + strategy tag is a Week-7+ thing. ADR 0013 pending.
-

10. Reference

10.1 HTTP endpoints

Method	Path	Purpose
GET	/healthz	Liveness
GET	/brochure/{crd}	List current Part 2A brochures for a CRD
POST	/pipeline/run	Submit a pipeline run; returns 202 + status URL
GET	/pipeline/run/{trace_id}	Poll pipeline state
POST	/report/decision	Write a CCO decision row
GET	/report/decision/{trace_id}	Read decision history

10.2 CLI commands

Command	Purpose
python -m adv_lens.ingestion.cli fetch-brochure <CRD>	Fetch one brochure
python -m adv_lens.ingestion.cli search <query>	IAPD search
python -m adv_lens.segmenter.cli <PDF>	Dump Item 1-18 sections
python -m adv_lens.app.graph.cli <CRD>	Run the full pipeline synchronously
python -m adv_lens.app.main	Start the FastAPI app
python -m adv_lens.app.jobs.reaper [--dry-run]	Sweep stuck pipeline_runs rows
python -m adv_lens.redline.cli <report.json> [--pdf]	Render a saved RedlineReport to HTML / PDF
python -m eval.runner [<section>]	Run the eval harness

10.3 Environment variables

The full list is in `.env.example`. Highlights:

Var	Default	Purpose
ANTHROPIC_API_KEY	—	Required for any LLM-backed run
MODEL_FEE_EXTRACTOR	claude-sonnet-4-6	Per-node override
MODEL_DISCIPLINARY	claude-haiku-4-5-20251001	Per-node override
MODEL_CONFLICTS	claude-sonnet-4-6	Per-node override
MODEL_REDLINE	claude-opus-4-7	Per-node override
POSTGRES_DSN	local docker-compose	Audit + pipeline_runs
QDRANT_URL	http://localhost:6333	Peer corpus
LANGFUSE_*	(blank until provisioned)	Trace export
ENABLE_HITL	true	When false, gate auto-approves (dev / batch mode)

Var	Default	Purpose
PIPELINE_RUN_REAP_THRESHOLD_MINUTES	10	Stuck-row reaper cutoff

10.4 File layout

src/adv_lens/	
app/	
main.py	# FastAPI app
graph/	
pipeline.py	# LangGraph topology
state.py	# ADVState
nodes/	# Per-node implementations
jobs/	
models.py	# PipelineRun
runner.py	# async background runner
reaper.py	# stuck-row sweeper
storage/	
audit.py	# HumanReview, llm_calls
db.py	# session + engine
extractors/	
fee.py	# Fee extractor
disciplinary.py	# Disciplinary extractor
conflicts.py	# Conflicts extractor
redline.py	# Redline writer
schemas.py	# All Pydantic models
segmenter/	# Item 1-18 segmenter
ingestion/	# IAPD + IARD clients
retrieval/	# Qdrant + hybrid search
llm/	# LLMClient + audit sink + cost
observability/	# Langfuse callbacks
eval/	
runner.py	# Eval entry point
scorers/	# Per-section scorers
fixtures/	# Golden set (19 today, 65 target)
results/<UTC>/	# Per-run reports
docs/	
architecture.md	# Pipeline + ADR index
compliance.md	# (placeholder)
user-manual.md	# This document
parking-lot.md	# Captured ideas
adr/	# 11 ADRs
tests/	# 221 pytest tests

10.5 ADR index

#	Title
0001	Stack choices
0002	Data sources (IAPD/IARD)
0003	Segmenter strategy
0004	Peer corpus indexing
0005	Extractor pattern
0006	Parallel state composition
0007	Hybrid retrieval (revises 0004 § 5)
0008	Redline output format
0009	LLM-as-judge for redline scoring (pending)
0010	HITL gate (marker-style)
0011	Async pipeline worker
0012	On-prem branch scope (pending)
0013	Peer auto-discovery from IARD (pending)

Each ADR follows the Context / Decision / Consequences format and is one file under `docs/adr/`.